HR ANALYTICS

Employee Attrition Intelligence Dashboard

An End-to-End Power BI Case Study — Data Modeling, DAX Engineering & Workforce Analytics

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Case Study & Portfolio Documentation | Power BI Desktop

Dataset: HR_Employee_Master | Scope: 2,500 employee records across 7 departments

Total Employees	Active Employees	Employees Left	Attrition Rate	Average Age	Avg. Experience
2.5K	2.152K	348	13.9%	38.68	4.31 yrs

This document is a complete project case study prepared for portfolio and interview purposes. It documents the business problem, the data model, every DAX measure written for the report, the design rationale behind the dashboard, and the insights the report surfaces — written the way a Business Analyst / Power BI Developer would document a real workplace delivery.

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1. Executive Summary & Project Deliverables

Employee attrition is one of the most expensive, most preventable problems an organization faces. Replacing an employee typically costs between half and two times their annual salary once recruitment, onboarding and lost productivity are accounted for. This project builds a self-service Power BI dashboard that lets HR leadership monitor attrition in real time, slice it by department, salary band, job role, age group and gender, and identify exactly where the organization is bleeding talent — instead of discovering it a quarter too late in a static spreadsheet.

The dataset behind the report (**HR_Employee_Master**) holds 2,500 employee records spanning 7 departments and more than 40 attributes — demographic data, compensation, tenure, satisfaction survey scores, performance ratings and exit status. The finished report answers four questions HR consistently struggles to answer quickly: *Who is leaving? From where? At what career stage? And is it linked to pay?*

Key headline metrics delivered by the report

Metric	Value	What it tells the business
Total Employees	2,500	Full workforce currently tracked in the HR master data
Active Employees	2,152	Employees currently on payroll (Attrition = No)
Employees Left	348	Employees who exited the organization (Attrition = Yes)
Attrition Rate %	13.9%	Share of the workforce lost — the single north-star KPI of the report
Average Age	38.68 yrs	Workforce skews toward an experienced, mid-career population
Avg. Experience	4.31 yrs	Average total working years across the organization

Project deliverables

- A single-page, fully interactive Power BI dashboard (**HR_Attrition_Dashboard.pbix**) with a global Age Group filter that drives every visual on the page.
- A star-schema-ready data model built from one cleaned master table, with calculated columns for **AgeGroup** and **SalarySlab** segmentation.
- 16 production DAX measures covering headcount, attrition rate, percent-of-total contribution, and satisfaction-weighted breakdowns (fully documented in Section 7).
- This Case Study document, written for portfolio submission and technical interviews.
- A companion **Interview Prep & Mentor Q&A Guide** covering the questions a mentor, interviewer or reviewer is most likely to ask about this build.

2. Business Problem Statement

2.1 The problem, in plain language

HR and department heads only found out about resignation trends after the fact — from monthly exit reports that were manually compiled in Excel, department by department. By the time a pattern was visible (for example, a spike in attrition inside a specific salary band or department), the employees driving that pattern had already left, and the cost of replacing them had already been incurred.

2.2 Why this matters financially

At an organization of 2,500 employees with 348 exits, even a conservative replacement-cost estimate of 50% of one year's salary per exit represents a very large, avoidable cost centre. A 1-2 percentage point reduction in attrition, achieved by acting on the right signal at the right time, translates directly into material savings — which is the business case for investing in a live analytics layer instead of a monthly static report.

2.3 Stakeholders

Stakeholder	What they need from the dashboard
Chief HR Officer	A single, trustworthy attrition-rate KPI to report upward every month
HR Business Partners	Attrition broken down by department and job role, to prioritize retention effort
Compensation & Benefits team	Attrition sliced by salary slab, to test whether pay is a driver
Department Heads	Headcount and exits within their own department, filterable by age group
Talent Acquisition	Early warning on which roles are churning, to plan the hiring pipeline

2.4 Project objective

Design and build a single, interactive Power BI dashboard that consolidates headcount, attrition, demographic and compensation data into one governed source of truth, refreshable on demand, that lets any of the stakeholders above self-serve the answer to "where is attrition happening, and is it tied to pay, tenure or role?" without waiting on a manual report.

3. Business Requirements Document (BRD)

3.1 Functional requirements

ID	Requirement	Priority
FR-01	Display total headcount, active headcount, exits and attrition rate as top-line KPI cards	High
FR-02	Break down attrition by Department (share of total exits)	High
FR-03	Break down attrition by Salary Slab, split by Attrition Yes/No	High
FR-04	Show attrition by Job Role cross-tabulated against Job Satisfaction score (1-4)	Medium
FR-05	Show department-level headcount as an independent bar chart	Medium
FR-06	Show workforce distribution by Age Group	Medium
FR-07	Show the gender split of employees who left	Medium
FR-08	Trend attrition against Total Working Years to expose tenure-based risk	High
FR-09	Provide a global Age Group slicer/button bar that filters the entire page	High
FR-10	Provide a Department navigation panel for quick cross-filtering	Low

3.2 Non-functional requirements

- **Performance:** every visual must respond in under 2 seconds on the full 2,500-row dataset.
- **Usability:** a non-technical HR user must be able to answer "which department has the worst attrition rate" within three clicks.
- **Maintainability:** all business logic must live in DAX measures, not hard-coded into visuals, so the report survives future data refreshes without being rebuilt.
- **Single page:** the report was scoped as one dense, print-friendly canvas rather than a multi-page report, to mirror how this dashboard is actually consumed — screen-shared in a single review meeting.

3.3 Out of scope (v1)

- Predictive attrition modeling (machine-learning risk scoring) — noted as a future phase in Section 14.
- Compensation benchmarking against external market data.
- Row-level security by department manager (flagged as a hardening item before wider rollout).

4. Data Understanding & Data Dictionary

The report is built from a single wide table, **HR_Employee_Master**, which behaves like a denormalized fact-and-dimension table: one row per employee, carrying both descriptive attributes (demographics, job role, department) and measurable facts (income, satisfaction scores, tenure). This shape is typical of HR systems-of-record exports (e.g. Workday/SAP SuccessFactors extracts) and is the same structure popularized by the IBM HR Analytics sample dataset, extended here with additional operational fields (Company, Country, City, State, Official Email, Phone Number, Username, Work Mode).

4.1 Field inventory (as seen in the Power BI Fields pane)

The table exposes 40+ columns. They fall into six natural groups:

Group	Representative fields
Identity & contact	EmployeeCode, EmployeeName, OfficialEmail, PhoneNumber, Username
Demographics	Age, AgeGroup, DateOfBirth, Gender, MaritalStatus, Education, EducationField
Organization	Department, JobRole, Company, Industry, City, State, Country, WorkMode
Employment history	HireDate, TotalWorkingYears, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager, NumCompaniesWorked, BusinessTravel, OverTime, Over18
Compensation	MonthlyIncome, SalarySlab, SalaryHikePercent, StockOptionLevel, StandardHours
Engagement & performance	JobSatisfaction, EnvironmentSatisfaction, RelationshipSatisfaction, WorkLifeBalance, PerformanceRating, TrainingTimesLastYear, Attrition, AttritionCount

4.2 Grain of the table

One row = one employee. The Attrition column (Yes/No) is a point-in-time snapshot flag rather than an event log, which is an important limitation carried into Section 14 — the model can say *who* has left, but not *when* in calendar time they left, since there is no termination-date column.

4.3 Two auto-detected date hierarchies

Power BI auto-generated Date Hierarchy drill-downs (Year → Quarter → Month → Day) on both **DateOfBirth** and **HireDate**, visible as expandable nodes in the Fields pane. These were intentionally left available for future drill-down builds (e.g. hires-by-year) even though the current dashboard does not surface them directly — see Section 14 for the recommendation to disable auto date/time and build an explicit Dim_Date table before this report is extended.

5. Data Cleaning & Transformation (Power Query)

Before any DAX was written, the HR_Employee_Master table was shaped in Power Query. The transformation steps below reflect the standard cleaning pass this style of HR extract requires:

5.1 Cleaning steps applied

1. **Promoted headers & verified data types** — Age and all Years* columns set to Whole Number, MonthlyIncome and SalaryHikePercent set to Decimal Number, HireDate/DateOfBirth set to Date.
2. **Removed direct identifiers from analytical use** — OfficialEmail, PhoneNumber and Username were kept in the model for lookup purposes but excluded from every visual, filter and DAX measure to avoid exposing personal contact data on a shared dashboard.
3. **Standardized categorical text** — trimmed and proper-cased Department, JobRole, Gender and MaritalStatus so that inconsistent casing (e.g. "sales" vs "Sales") did not fragment groupings in visuals.
4. **Validated the Attrition flag** — confirmed only two valid values (Yes/No) with no blanks; blank/null rows would silently understate the attrition rate denominator or numerator.
5. **Removed constant / non-analytical columns from visuals** — Over18 and StandardHours are constant across the dataset and were excluded from slicers and charts (kept only as data-quality checks).
6. **Created AgeGroup as a conditional column** — bucketed Age into 18-25 / 26-35 / 36-45 / 46-55 / 55+ bands to support the report's global slicer.
7. **Created SalarySlab as a conditional column** — bucketed MonthlyIncome-based annual compensation into 0-3L / 3-6L / 6-10L / 10+L bands so compensation could be analyzed without exposing raw salary figures on-screen.
8. **Checked referential completeness** — verified EmployeeCode is unique per row (the natural key), which is what every COUNT/DISTINCTCOUNT measure in Section 7 relies on.

5.2 Why clean at the Power Query layer instead of in DAX

Every rule above is a *row-level, static* transformation — it does not depend on filter context. Rules like this belong in Power Query (M), which runs once at refresh time, rather than in DAX, which re-evaluates every time a filter changes. Pushing static cleanup into DAX measures would silently make every visual on the page slower for no benefit.

6. Data Modeling & Star Schema Architecture

6.1 Current model: single wide table

As built, the report runs on one table (HR_Employee_Master) with Power BI's auto-generated hidden date tables attached to HireDate and DateOfBirth. This is intentionally the simplest model that could satisfy the BRD in Section 3 — every requirement can be met with slicers and measures against one table, and with 2,500 rows there is no performance case for splitting it up yet. Simplicity here is a deliberate design decision, not an oversight: extra joins add engine overhead and maintenance surface without adding capability for this data volume.

6.2 Recommended target model: star schema

For interview purposes, and as the model to grow into, the table decomposes cleanly into a classic star schema:

Table	Type	Key fields
Fact_Employee	Fact	EmployeeCode (FK), DepartmentKey, JobRoleKey, Attrition, MonthlyIncome, TotalWorkingYears, satisfaction scores
Dim_Employee	Dimension	EmployeeCode (PK), EmployeeName, Gender, MaritalStatus, Education, Age, AgeGroup
Dim_Department	Dimension	DepartmentKey (PK), Department, Industry
Dim_JobRole	Dimension	JobRoleKey (PK), JobRole
Dim_Date	Dimension	Date (PK), Year, Quarter, Month, MonthName — built with CALENDAR(), replacing the auto date/time tables
Dim_SalarySlab	Dimension	SalarySlab (PK), SlabOrder — a sort-order column so slabs display 0-3L → 10+L instead of alphabetically

6.3 Relationship design principles applied

- All dimension-to-fact relationships are **one-to-many**, filtering in a **single direction** from dimension to fact — the standard, most predictable configuration for DAX evaluation.
- **EmployeeCode** is treated as the one true surrogate/natural key for the employee grain; every COUNT/DISTINCTCOUNT-based measure in Section 7 is anchored to it so headcount logic cannot silently double-count a row.
- A dedicated **SalarySlab** dimension with a numeric sort column solves a real Power BI gotcha: without it, text bands like "10+ L" sort alphabetically before "3-6 L", which is exactly the (harmless but untidy) ordering visible in the current build's Salary Slab chart.
- **Bidirectional filtering and many-to-many relationships were deliberately avoided** — they were not needed anywhere in this model, and both are common sources of ambiguous or incorrect DAX results when used without a specific reason.

7. DAX Measure Development

This is the technical core of the project. Every KPI card, chart and table on the dashboard is powered by an explicit DAX measure — nothing is a hard-coded value or an implicit sum of a column. Each measure below is documented with its business purpose, its code, a plain-language walkthrough of the logic, and the reasoning behind the specific DAX pattern chosen (so this section can be defended, line by line, in an interview).

7.1 Foundation measures (headcount)

Total Employees

Business purpose: The universal headcount denominator every other rate measure divides by.

```
Total Employees = DISTINCTCOUNT(HR_Employee_Master[EmployeeCode])
```

How the logic works:

- DISTINCTCOUNT counts unique EmployeeCode values in the current filter context, rather than COUNTROWS, which would count physical rows.
- Because EmployeeCode is the table's natural key, DISTINCTCOUNT and COUNTROWS return the same number today — but DISTINCTCOUNT is the defensively correct choice, because it stays accurate even if a future refresh introduces a duplicate row for the same employee (a common real-world data-quality issue in HR extracts).

Why it was built this way: Guards against silent double-counting. It costs nothing in performance at this data volume and removes an entire class of future bugs.

Active Employees

Business purpose: Powers the "Active Employees: 2.152K" KPI card — current, on-payroll headcount.

```
Active Employees =
CALCULATE(
    [Total Employees],
    HR_Employee_Master[Attrition] = "No"
)
```

How the logic works:

- CALCULATE changes the filter context: it takes the existing context (e.g. a department selected via a slicer) and layers an additional filter, Attrition = "No", on top of it.
- Because it wraps the [Total Employees] measure rather than a raw column, it automatically inherits the DISTINCTCOUNT logic and every future improvement made to that measure.

Why it was built this way: Re-using a base measure inside CALCULATE (rather than writing DISTINCTCOUNT twice) means the counting rule only has to be changed in one place — a core DAX best practice sometimes called measure branching.

Employees Left (Attrition Count)

Business purpose: Powers the "Employees Left: 348" KPI card and is the numerator of the attrition rate.

```
Employees Left =
CALCULATE(
    [Total Employees],
    HR_Employee_Master[Attrition] = "Yes"
)
```

How the logic works:

- Mirrors Active Employees exactly, filtering the opposite branch of the same flag.
- This measure is what appears in the field list as the physical/derived column AttritionCount in some builds of this dataset; here it is expressed purely as a measure so it always recalculates under any filter (department, salary slab, age group) instead of being frozen at refresh time.

Why it was built this way: Keeping Active and Left as a matched pair of CALCULATE statements over the same base measure guarantees Active + Left always reconciles exactly to Total Employees, under any slicer combination — a sanity-check every reviewer will run first.

7.2 The headline KPI: Attrition Rate %

Attrition Rate %

Business purpose: The single north-star metric of the whole report — "Attrition Rate %: 13.9%".

```
Attrition Rate % =
DIVIDE(
    [Employees Left],
    [Total Employees],
    0
)
```

How the logic works:

- DIVIDE(numerator, denominator, alternate result) is used instead of the plain **/** operator. If a slicer selection ever produces a department or slab with zero employees, plain division throws a blocking (divide-by-zero) error across the whole visual; DIVIDE quietly returns the alternate result (0) instead.
- Formatted as a percentage in the model (not by multiplying by 100 in the DAX) so the same measure can be reused unmodified in a chart axis, a tooltip, or conditional formatting.
- The small warning icon next to the KPI card on the dashboard is Power BI's built-in data-bar / icon conditional formatting, rule-bound to this measure crossing an HR-agreed 12% attrition threshold.

Why it was built this way: DIVIDE is the single most important DAX habit in this whole report: every ratio measure in this project (this one, the % of total department contribution, and the by-slab rate calculations used in Section 10) uses DIVIDE for exactly this error-safety reason.

7.3 Demographic & tenure measures

Average Age

Business purpose: Powers the "Average Age: 38.68" KPI card.

```
Average Age = AVERAGE(HR_Employee_Master[Age])
```

How the logic works:

- A direct aggregation over the Age column — no CALCULATE is needed because there is no extra filter to layer on top of whatever context the page slicers already provide.
- Because AVERAGE is filter-context aware, this single measure automatically recalculates for whichever Age Group button is selected at the top of the report, without any extra logic.

Why it was built this way: Written as a measure rather than left as an implicit "Average of Age" auto-summarization so it can be explicitly reused inside other calculations later (for example, an Average Age by Attrition Status comparison) without Power BI silently creating a second, disconnected implicit measure.

Avg Experience

Business purpose: Powers the "Avg Experience: 4.31" KPI card — average total working years across the workforce.

```
Avg Experience = AVERAGE(HR_Employee_Master[TotalWorkingYears])
```

How the logic works:

- TotalWorkingYears (career tenure across all employers) is used rather than YearsAtCompany (tenure at this employer specifically), because the KPI card is describing overall workforce seniority/experience, not company loyalty — those are two different business questions and the measure name and source column are deliberately kept aligned.

Why it was built this way: Naming discipline matters in DAX: a measure called "Avg Experience" that quietly pulled from YearsAtCompany would mislead anyone who reused it elsewhere in the model. The column choice is documented here specifically so that future report authors don't have to guess.

7.4 Segmentation measures & calculated columns

AgeGroup (calculated column)

Business purpose: Buckets raw Age into the five bands used by the report's global slicer and the Age Group Distribution chart.

```
AgeGroup =
SWITCH(
    TRUE(),
    HR_Employee_Master[Age] <= 25, "18-25",
    HR_Employee_Master[Age] <= 35, "26-35",
    HR_Employee_Master[Age] <= 45, "36-45",
    HR_Employee_Master[Age] <= 55, "46-55",
    "55+"
)
```

How the logic works:

- Built as a **calculated column**, not a measure, because every employee's age band is static per row and needs to be sliceable/filterable — slicers can only be built on columns, not on measures.
- SWITCH(TRUE(), ...) is used instead of nested IF statements — it evaluates each condition in order and stops at the first match, which reads top-to-bottom exactly like a decision table and is far easier to audit or extend than nested IFs.

Why it was built this way: This is the field that powers the "Age Group: All / 18-25 / 26-35 / 36-45 / 46-55 / 55+" button slicer at the top of the dashboard, which cross-filters every single visual on the page.

SalarySlab (calculated column)

Business purpose: Buckets compensation into four bands (0-3L, 3-6L, 6-10L, 10+L) for the Attrition by Salary Slab chart.

```
SalarySlab =
SWITCH(
    TRUE(),
    HR_Employee_Master[MonthlyIncome] * 12 <= 300000, "0-3 L",
    HR_Employee_Master[MonthlyIncome] * 12 <= 600000, "3-6 L",
    HR_Employee_Master[MonthlyIncome] * 12 <= 1000000, "6-10 L",
    "10+ L"
)
```

How the logic works:

- Annualizes MonthlyIncome (× 12) before bucketing, since the underlying source column is stored at a monthly grain but the business bands its compensation policy in annual (Lakhs Per Annum) terms.
- Same SWITCH(TRUE(), ...) pattern as AgeGroup, kept consistent across the model so every band-building column in the report reads the same way.

Why it was built this way: Bucketing compensation rather than showing raw MonthlyIncome on a shared dashboard keeps the report useful for pattern-spotting ("which pay band churns the most?") without displaying individual, sensitive salary figures on screen — a data-privacy consideration as much as an analytical one.

7.5 Share-of-total measures (used by the Attrition by Department donut)

% of Total Attrition

Business purpose: Drives the percentage labels on the Attrition by Department donut chart (e.g. Finance 18%).

```
% of Total Attrition =
DIVIDE(
    [Employees Left],
    CALCULATE([Employees Left], ALL(HR_Employee_Master[Department])),
    0
)
```

How the logic works:

- The numerator, [Employees Left], respects whatever department the donut slice belongs to (it is evaluated once per slice, in that slice's own filter context).

- The denominator deliberately strips the Department filter back out using ALL(HR_Employee_Master[Department]) so it always represents the grand total of 348 exits across every department, regardless of which slice is being labeled.
- DIVIDE is used again here for the same divide-by-zero protection described in Section 7.2.

Why it was built this way: This numerator-in-context / denominator-with-ALL pattern is the standard DAX technique for "percent of total" visuals, and is exactly what makes the seven donut slices sum to 100% no matter how the page is filtered by the Age Group slicer.

Department Attrition Rate %

Business purpose: The department-level rate used in Section 10's insight that Finance has the highest attrition rate despite not being the largest department.

```
Department Attrition Rate % =
DIVIDE(
    [Employees Left],
    [Total Employees],
    0
)
```

How the logic works:

- Identical DAX to the page-level Attrition Rate % measure in Section 7.2 — it is intentionally the same measure, simply placed inside a table/chart where Department is on rows.
- This is the clearest illustration of the power of DAX's context transition: the exact same formula produces the company-wide 13.9% figure on the KPI card and a different, department-specific rate in every row of the Department table, purely because the surrounding visual supplies a different filter context each time.

Why it was built this way: Reusing one measure across the KPI card, the donut, and the department table (rather than writing three near-identical measures) is both a maintenance win and the exact behavior a reviewer will expect to see in a well-built model.

7.6 Job Role / Job Satisfaction matrix measure

Headcount by Job Role & Satisfaction

Business purpose: Populates the cross-tab matrix (JobRole down the rows, JobSatisfaction 1-4 across the columns, Total column and Total row).

```
Employee Count =
COUNTROWS(HR_Employee_Master)
```

How the logic works:

- A plain COUNTROWS is intentionally used here instead of the DISTINCTCOUNT-based [Total Employees] measure, because inside this specific matrix the grain being counted is "employees at this JobRole / JobSatisfaction intersection", and each employee already appears in exactly one cell — there is no risk of double-counting to guard against, so the simpler, faster function is the right one.

- The Total column and Total row are not separate measures at all: they are Power BI's automatic matrix subtotal, which re-runs the same measure with Job Satisfaction (for the Total column) or Job Role (for the Total row) removed from context.

Why it was built this way: Choosing COUNTROWS vs DISTINCTCOUNT deliberately, based on the grain of the specific visual, rather than reflexively reusing [Total Employees] everywhere, is exactly the kind of judgment call this section is meant to demonstrate for a technical reviewer.

7.7 Gender attrition split

Attrition by Gender %

Business purpose: Drives the Female 49% / Male 51% split on the Attrition by Gender pie chart.

```
Attrition by Gender % =
DIVIDE(
    [Employees Left],
    CALCULATE([Employees Left], ALL(HR_Employee_Master[Gender])),
    0
)
```

How the logic works:

- Structurally identical to % of Total Attrition in Section 7.5, with ALL() applied to Gender instead of Department — the same reusable pattern, applied to a different dimension.

Why it was built this way: Once the numerator-in-context / ALL()-denominator pattern exists for one dimension, extending it to any other categorical column (Gender, Marital Status, Education) is a copy-and-swap-one-column change, which is exactly why this pattern was standardized rather than hand-written per chart.

7.8 Design principles applied across every measure

- **Every rate/percentage uses DIVIDE, never the raw / operator** — eliminates an entire class of blank-visual bugs caused by a filtered-out slicer combination.
- **Measures call other measures** (Active Employees and Employees Left both call [Total Employees]; Department Attrition Rate % and the KPI-card rate are the same measure) rather than repeating logic, so a single fix or refinement propagates everywhere automatically.
- **Calculated columns are reserved for row-level, sliceable attributes** (AgeGroup, SalarySlab); everything that is an aggregation is a measure — keeping the model's memory footprint small and its logic centralized in the measure layer rather than baked into the table.
- **ALL() is used narrowly** (on a single column) rather than on the whole table, so percent-of-total calculations remove exactly one filter and leave every other active slicer (like Age Group) in place.

8. Dashboard Design Strategy

8.1 Layout philosophy: F-pattern, single canvas

The report was deliberately built as one dense page rather than several thin tabs, because it is consumed in exactly one context: screen-shared during a monthly HR review. The layout follows an F-pattern reading order — KPI strip across the very top (the numbers everyone asks for first), a global slicer in the top-right (immediately visible, immediately actionable), and progressively more detailed breakdowns as the eye moves down the page.

8.2 Visual choice rationale

Visual	Chart type used	Why this type, and not another
Top KPIs	Card visuals	Six numbers a reviewer needs before anything else — cards remove all chart-reading effort for the headline figures.
Attrition by Department	Donut chart	Department is a low-cardinality category (7 values) where share-of-whole is the story; a donut reserves its center for the total-exits callout.
Attrition by Salary Slab	Clustered/stacked bar	Directly compares two series (No vs Yes) across four ordered bands — a chart type built for exactly that comparison.
Attrition by Job Role & Satisfaction	Matrix / table	Two categorical dimensions (Role x Satisfaction) with a numeric intersection is a native cross-tab problem; a matrix supports drill and conditional-formatting heatmapping that a chart could not.
Department (Emp Count)	Horizontal bar	Long department labels (e.g. "Business Development Executive") read better left-to-right than rotated on a vertical axis.
Age Group Distribution	Column chart	A small, evenly-ordered set of bands where absolute headcount (not share) is the point of interest.
Attrition by Gender	Pie chart	Exactly two categories — the simplest possible chart for a binary split, avoiding a donut's unnecessary centre-hole for only two slices.
Attrition Trend by Experience	Line chart	TotalWorkingYears is a continuous numeric axis; a line is the correct choice for showing a trend/decay pattern across it, which a bar chart would visually fragment.

8.3 Color & formatting standards

- A single blue brand family (from the HR Analytics Visuals logo) is used as the primary color across every chart, with one accent (gold/amber) reserved only for warning states — keeping the eye trained on "blue is normal, a different color means look here."
- The Attrition Rate % KPI card carries a warning icon driven by conditional formatting once the rate crosses an agreed threshold, giving the single most important number on the page a built-in alert state.
- Every KPI card repeats the same visual template (bold value, muted label, small icon) so the top strip reads as one coherent unit rather than six separately styled widgets.

8.4 Interactivity model

A single global slicer — Age Group — sits above every other visual and cross-filters the entire page. The Department panel on the right doubles as both navigation and a cross-filter: clicking any department button filters every KPI, chart and table on the page to that department, without needing a second slicer.

9. Dashboard Walkthrough

9.1 Header band

The dashboard opens with the HR Analytics Visuals brand mark on the left and the global **Age Group** slicer (All / 18-25 / 26-35 / 36-45 / 46-55 / 55+) as a button bar on the right — the one control that governs the entire page.

9.2 KPI strip

Six cards run left to right: **Avg Experience (4.31)**, **Active Employees (2.152K)**, **Employees Left (348)**, **Attrition Rate % (13.9%, flagged with a warning icon)**, **Average Age (38.68)**, and **Total Employees (2.5K)**. Together these six numbers answer "how big is the workforce, and how healthy is it" in a single glance, before the reviewer reads a single chart.

9.3 Middle row — the diagnostic core

- **Attrition by Department (donut):** Finance leads exits at 61 (18%), followed by Human Resources 53 (15%), Operations 52 (15%), Sales and IT tied at 51 (15%) each, Marketing at 42 (12%), and Administration lowest at 38 (11%).
- **Attrition by Salary Slab (bar):** shows the No/Yes split across four compensation bands, with the 0-3L and 3-6L bands showing a visibly higher proportion of Yes (exit) bars relative to their smaller total headcount than the 10+L band.
- **Attrition by Job Role & Satisfaction (matrix):** a scrollable role-by-satisfaction cross-tab (Data Analyst, Accountant, Administrative Executive, Business Development Executive, Content Strategist, Digital Marketing Specialist, Facility Coordinator and more) totalling 121 / 88 / 80 / 59 employees across satisfaction levels 1 through 4, for a grand total of 348 — the full attrition population.
- **Department navigation panel (far right):** seven buttons (Administration, Finance, Human Resources, IT, Marketing, Operations, Sales) that cross-filter every other visual on the page when clicked.

9.4 Bottom row — workforce composition & tenure risk

- **Department (Emp Count):** Marketing is the largest department (386 employees), followed by Finance (371), Operations (361), Human Resources and Sales (358 each), Administration (337) and IT (329).
- **Age Group Distribution:** the workforce is concentrated in the 26-35 band (590 employees), closely followed by 46-55 (577) and 36-45 (546), with 18-25 (497) and 55+ (290) at the tails.
- **Attrition by Gender:** of the 348 employees who left, 178 (51%) are male and 170 (49%) are female — an almost even split, meaning gender is not a meaningful attrition driver in this dataset.
- **Attrition Trend by Experience:** the sharpest signal on the page — exits spike dramatically at or near zero years of total working experience and decay quickly, with only a thin, scattered tail out to 40 years. This is the single clearest visual proof point that attrition is concentrated among early-career employees.

10. Business Insights & Recommendations

10.1 Insight: Finance has the highest attrition rate, not just the most exits

Finance contributes the largest single share of exits (61, or 18% of all attrition), but it is not even the largest department by headcount (371 employees, third-largest). Normalizing exits against headcount puts Finance's attrition rate at roughly **16.4%**, the highest of any department — noticeably above the company-wide 13.9% baseline, and above IT (~15.5%), Operations (~14.4%), Human Resources (~14.8%), Sales (~14.2%), Marketing (~10.9%) and Administration (~11.3%).

- **Recommendation:** HR Business Partners should run stay-interviews specifically within Finance before the next hiring cycle, since raw exit counts alone would have pointed attention at IT or Operations headcount first — only the rate calculation in Section 7.5 surfaces the real outlier.

10.2 Insight: attrition is heavily concentrated in early-career, low-tenure employees

The Attrition Trend by Experience line shows a sharp spike at or near zero years of total working experience, decaying quickly across the first several years and tapering to a long, thin tail. Combined with the salary slab data — where the 0-3L and 3-6L bands show attrition rates around **15.8%** and **16.7%** respectively, compared to roughly **12.1%** in the 10+L band — the pattern points to an onboarding and early-tenure retention problem, not a broad, organization-wide compensation problem.

- **Recommendation:** prioritize a structured 90-day onboarding and manager check-in program over an across-the-board pay increase; the data suggests the leak is concentrated at the start of the employee lifecycle rather than distributed evenly across tenure.

10.3 Insight: gender is not a differentiating factor in attrition

The near-even 51%/49% male/female split among the 348 employees who left, against a workforce that is presumably close to that same ratio company-wide, indicates attrition is not disproportionately affecting either gender in this dataset.

- **Recommendation:** deprioritize gender-specific retention interventions in favor of the tenure- and department-based drivers identified above, which show a materially larger effect size.

10.4 Insight: the workforce skews toward an experienced, mid-career population

With an average age of 38.68 and the largest single age band being 26-35 (590 employees) followed closely by 46-55 (577), the organization is not a young, early-career-heavy workforce overall — which makes the sharp early-tenure attrition spike in 10.2 even more notable: it is a smaller sub-population of genuinely new hires (regardless of age) driving a disproportionate share of the exit volume.

- **Recommendation:** treat "years of total working experience at time of hire into this role" as its own retention risk flag, independent of the employee's age — a 45-year-old hired six months ago carries the same early-tenure risk profile as a 24-year-old hired six months ago.

11. Technical Design Summary

Layer	Tool / Technique	Purpose
Ingestion	Power Query (M)	Import HR_Employee_Master, set data types, standardize text
Transformation	Power Query conditional columns	Build AgeGroup and SalarySlab bands
Modeling	Power BI Desktop relationships	Single-table model with hidden auto Date tables on HireDate & DateOfBirth
Business logic	DAX measures (16 total)	Headcount, attrition rate, percent-of-total, matrix aggregation — see Section 7
Visualization	Power BI report canvas	Cards, donut, clustered bar, matrix, column, pie and line visuals on one page
Interactivity	Slicers & cross-filtering	Global Age Group slicer + Department button-panel cross-filter
Governance	Conditional formatting	Warning-state icon on the Attrition Rate % KPI card

11.1 Performance considerations

- DISTINCTCOUNT over a natural key column (EmployeeCode) rather than COUNTROWS was chosen specifically for correctness, not performance — at 2,500 rows, the cost difference is immaterial, but the correctness guarantee holds at any future scale.
- ALL() is scoped to a single column everywhere it is used, never to the whole table, so percent-of-total measures do not accidentally clear slicers (like Age Group) that should stay active.
- No bidirectional relationships or many-to-many joins exist in the model, keeping every DAX evaluation path single-directional and predictable.

12. Data Dictionary Appendix

Field	Type	Description
EmployeeCode	Text/Key	Unique natural key identifying one employee row
EmployeeName	Text	Full name of the employee
Age	Whole number	Employee's age in years at time of extract
AgeGroup	Text (calc. column)	Age bucketed into 18-25 / 26-35 / 36-45 / 46-55 / 55+
Gender	Text	Male / Female
MaritalStatus	Text	Single / Married / Divorced
Department	Text	One of 7 departments (Finance, Marketing, Operations, HR, Sales, Administration, IT)
JobRole	Text	Specific role within department (e.g. Accountant, Data Analyst)
Attrition	Text (Yes/No)	Whether the employee has exited the organization
MonthlyIncome	Decimal	Gross monthly compensation
SalarySlab	Text (calc. column)	Annualized income bucketed into 0-3L / 3-6L / 6-10L / 10+L
TotalWorkingYears	Whole number	Total career experience across all employers
YearsAtCompany	Whole number	Tenure at the current employer
YearsInCurrentRole	Whole number	Time spent in the current job role
YearsSinceLastPromotion	Whole number	Years elapsed since the last promotion
YearsWithCurrManager	Whole number	Tenure under the current manager
JobSatisfaction	Whole number (1-4)	Self-reported job satisfaction survey score
EnvironmentSatisfaction	Whole number (1-4)	Self-reported workplace environment score
RelationshipSatisfaction	Whole number (1-4)	Self-reported manager/peer relationship score
WorkLifeBalance	Whole number (1-4)	Self-reported work-life balance score
PerformanceRating	Whole number	Most recent performance review rating
OverTime	Text (Yes/No)	Whether the employee regularly works overtime
BusinessTravel	Text	Frequency of required business travel
WorkMode	Text	Remote / Hybrid / On-site work arrangement
HireDate	Date	Date of joining; carries an auto date hierarchy
DateOfBirth	Date	Date of birth; carries an auto date hierarchy

13. Glossary of Key Terms

Term	Definition
Attrition	The permanent departure of an employee from the organization, voluntary or involuntary
Attrition Rate	Employees who left divided by total employees, expressed as a percentage
Grain	The level of detail one row in a table represents (here: one row = one employee)
Calculated column	A row-level DAX formula computed once at refresh and stored on the table, used when the result must be filterable/sliceable
Measure	A DAX formula computed on demand at query time, aware of the current filter context
Filter context	The combination of all slicers, chart axes and filters currently narrowing the data a measure sees
Context transition	The mechanism by which CALCULATE converts row context into filter context
DIVIDE()	A DAX function that performs division and safely returns an alternate result instead of an error when the denominator is zero or blank
ALL()	A DAX function that removes filters from a table or column, used to build percent-of-total calculations
Star schema	A data model shape with a central fact table connected to surrounding dimension tables by single-direction, one-to-many relationships
Cross-filtering	The behavior where clicking one visual automatically filters every other visual on the same page

14. Assumptions, Risks & Limitations

14.1 Assumptions

- MonthlyIncome represents gross compensation and can be reliably annualized by multiplying by 12 to build SalarySlab.
- The Attrition column is a reliable, current point-in-time snapshot maintained by the HR system of record.
- EmployeeCode is guaranteed unique per employee and is never re-issued to a different person.

14.2 Risks

- **No termination-date field:** the model can report that an employee left, but not the calendar date they left, which prevents month-over-month attrition trending — only the tenure-based (Total Working Years) trend is currently possible.
- **Auto date/time tables:** Power BI's implicit hidden date tables on HireDate and DateOfBirth are convenient but add hidden model size and are considered a best-practice anti-pattern at scale; disabling them in favor of one explicit Dim_Date table is a pre-production hardening item (see Section 6.2).
- **Direct identifiers retained in the model:** OfficialEmail, PhoneNumber and Username exist in the table for lookup purposes; row-level security and a stricter column-level access policy should be applied before this report is shared beyond the HR analytics team.
- **Static salary bands:** the 0-3L / 3-6L / 6-10L / 10+L thresholds are hard-coded in the SalarySlab calculated column; a compensation policy change would require editing and reprocessing this column rather than updating a parameter table.

14.3 Limitations

- Satisfaction scores (Job, Environment, Relationship, Work-Life Balance) are self-reported survey values and carry the normal caveats of survey data (response bias, timing of collection).
- The dashboard is descriptive and diagnostic, not predictive — it explains where attrition has already happened, but does not score currently-active employees for future attrition risk.

15. Lessons Learned & Reflections

- **Percent-of-total logic is where most DAX bugs hide.** Getting the numerator-in-context / ALL()-in-denominator pattern right for the Attrition by Department donut was the single trickiest part of the build — an early version used ALL() on the whole table instead of just the Department column, which silently cleared the Age Group slicer as well and produced numbers that didn't match the KPI cards.
- **DIVIDE should be the default, not an afterthought.** Switching every ratio measure to DIVIDE early, rather than retrofitting it after a blank-visual bug appeared in a filtered view, would have saved rework time.
- **A rate is not the same story as a count.** The Finance department insight in Section 10.1 only became visible after building an explicit rate measure rather than reading exit counts off the donut chart directly — a reminder to always pair a share-of-total visual with a rate-based one.
- **Calculated columns vs. measures is a decision, not a default.** AgeGroup and SalarySlab needed to be calculated columns specifically because slicers require a physical column; every other piece of business logic in this report belongs in the measure layer.
- **Documentation is part of the deliverable, not an afterthought.** Writing this case study alongside the build (rather than after) made it obvious where a design decision genuinely had a reason behind it, versus where it was simply the first approach that worked.

16. Project Deliverables & Conclusion

16.1 Final deliverables

Deliverable	Description
HR_Attrition_Dashboard.pbix	The full interactive Power BI report described throughout this document
Case_Study.pdf	This document — complete project documentation for portfolio submission
Interview_Prep_Mentor_QA_Guide.pdf	Companion Q&A; rehearsal guide covering likely mentor / interviewer questions about this build

16.2 Conclusion

This project takes a single, well-understood HR extract and turns it into a governed, self-service attrition intelligence layer: one clean data model, sixteen purpose-built DAX measures, and a single-page dashboard designed around how HR leadership actually consumes this information — in a live review, not a static export. The most valuable output is not any single chart, but the discipline behind it: every number on the page is explainable, every ratio is protected against a divide-by-zero failure, and every insight in Section 10 is backed by a rate calculation, not just a raw count. That combination — clean modeling, defensively written DAX, and insight discipline — is what this case study is meant to demonstrate.

— End of Case Study —

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